Elementary Analysis

Section 33: Properties of the Riemann Integral

Exercises 33.1–33.15, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Contents

33 Properties of the Riemann integral **3**

33.1 Exercise 33.1: Decreasing functions are integrable 3
33.2 Exercise 33.2: Scaling suprema and infima 3
33.3 Exercise 33.3: Step functions and postage 3
33.4 Exercise 33.4: An integrable absolute value does not suffice 4
33.5 Exercise 33.5: Bounding an integral without evaluating it 4
33.6 Exercise 33.6: Absolute value does not increase oscillation 4
33.7 Exercise 33.7: Squaring preserves integrability 5
33.8 Exercise 33.8: Products, maxima, and minima 5
33.9 Exercise 33.9: Uniform limits of integrable functions 6
33.10 Exercise 33.10: An oscillatory function that is still integrable 6
33.11 Exercise 33.11: An integrable function beyond the piecewise criteria 7
33.12 Exercise 33.12: The rational-denominator function has zero integral 7
33.13 Exercise 33.13: Equal integrals force an interior intersection 8

33.14 Exercise 33.14: A weighted mean-value theorem 8
33.15 Exercise 33.15: Narrow triangular peaks 9

33 Properties of the Riemann integral

The Darboux criterion converts integrability into an estimate of the total oscillation of a function over short intervals. Once integrability has been established, linearity, comparison, and interval additivity make calculations much easier. Keep two issues separate: first justify that an integral exists; then use its algebraic properties. A pointwise limit alone does not usually allow us to move a limit through an integral.

33.1 Exercise 33.1: Decreasing functions are integrable

What we must show. Complete the monotone-function integrability proof. Let f be decreasing on $[a, b]$, meaning nonincreasing. It is bounded between $f(b)$ and $f(a)$. If these values agree, f is constant and every upper and lower sum agrees. Otherwise $D = f(a) - f(b) > 0$. For a partition $P = \{t_0, \dots, t_n\}$, decreasingness gives

$$U(f, P) - L(f, P) = \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)](t_j - t_{j-1}) \le \operatorname{mesh}(P) \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)].$$

The last sum telescopes to D. Given $\varepsilon > 0$, choose an equal partition with mesh less than ε/D. Then the gap is below ε.

Conclusion. The Darboux criterion proves integrability, including the constant case.

33.2 Exercise 33.2: Scaling suprema and infima

What we must show. For a nonempty bounded $S \subseteq \mathbb{R}$ and $c > 0$, prove $\sup(cS) = c \sup S$ and $\inf(cS) = c \inf S$.

Prove the supremum statement using both defining properties.

Write $M = \sup S$. Every $s \in S$ satisfies $s \le M$; multiplication by the positive number c gives $cs \le cM$. Thus cM is an upper bound for cS. To show it is the least one, let $\varepsilon > 0$. Since $M - \varepsilon/c < M$, it is not an upper bound for S, so there is $s \in S$ with $M - \varepsilon/c < s \le M$. Multiplying gives $cM - \varepsilon < cs \le cM$. Hence no number below cM can be an upper bound for cS, proving the desired equality.

Repeat with lower bounds.

If $m = \inf S$, then $m \le s$ implies $cm \le cs$, so cm is a lower bound for cS. Given $\varepsilon > 0$, the number $m + \varepsilon/c$ is not a lower bound for S. Thus some $s \in S$ satisfies $m \le s < m + \varepsilon/c$, and $cm \le cs < cm + \varepsilon$. No number greater than cm is a lower bound for cS.

Conclusion. Both scaling identities follow; positivity of c was needed to preserve the inequality directions.

33.3 Exercise 33.3: Step functions and postage

What we must show. Integrate a finite step function and the postage-stamp example.

(a) Work one subinterval at a time.

Suppose $f = c_j$ on (u_{j-1}, u_j) for the finite partition in the exercise. On the closed interval $[u_{j-1}, u_j]$, the function differs from the constant c_j at most at its two endpoints. Exercise 32.7 therefore proves integrability there and gives integral $c_j(u_j - u_{j-1})$. Interval additivity, applied successively to the finitely many intervals, proves

$$\int_a^b f(x)\,dx = \sum_{j=1}^{m} c_j(u_j - u_{j-1}).$$

There is no condition on the finitely many partition-point values, except that they are real numbers.

(b) Apply the formula to the postage function.

On the four open unit intervals between 0 and 4, the values of P are $A, A+B, A+2B, A+3B$. Each interval has length 1; the value at 4 does not change the integral. Consequently

$$\int_0^4 P(x)\,dx = A + (A+B) + (A+2B) + (A+3B) = 4A + 6B.$$

Conclusion. Step functions are integrable, with integral equal to the sum of height times width; the postage integral is $4A + 6B$.

33.4 Exercise 33.4: An integrable absolute value does not suffice

What we must show. Find a nonintegrable f on $[0, 1]$ with integrable $|f|$. Set $f(x) = 1$ at rational points and $f(x) = -1$ at irrational points. Every nondegenerate interval contains both sorts of points. Thus every upper sum is 1 and every lower sum is -1; the two Darboux integrals differ, so f is not integrable. However, $|f(x)| = 1$ for every x, and a constant function is integrable, with integral 1 on $[0, 1]$.

Conclusion. This example shows that the converse of integrability of f implying integrability of $|f|$ is false.

33.5 Exercise 33.5: Bounding an integral without evaluating it

What we must show. Bound the absolute integral of $x^2 \sin^8(e^x)$ on $[-2\pi, 2\pi]$. The integrand is continuous, hence integrable. Since $0 \le \sin^8(e^x) \le 1$, we have $|x^2 \sin^8(e^x)| \le x^2$. The absolute-integral inequality and comparison give

$$\left|\int_{-2\pi}^{2\pi} x^2 \sin^8(e^x)\,dx\right| \le \int_{-2\pi}^{2\pi} x^2\,dx = 2\frac{(2\pi)^3}{3} = \frac{16\pi^3}{3}.$$

The integral of the square follows from the Darboux calculation in Section 32 (or, once established, the Fundamental Theorem).

Conclusion. The requested bound is $16\pi^3/3$.

33.6 Exercise 33.6: Absolute value does not increase oscillation

What we must show. Show, for a nonempty set S on which f is bounded,

$$\sup_S |f| - \inf_S |f| \le \sup_S f - \inf_S f.$$

Compare two values before taking extrema.

For arbitrary $x, y \in S$, the reverse triangle inequality gives

$$|f(x)| - |f(y)| \le |f(x) - f(y)| \le M - m, \quad M = \sup_S f, \quad m = \inf_S f.$$

The last inequality holds because both $f(x)$ and $f(y)$ lie in $[m, M]$. Fix y and take the supremum over x. This yields $\sup_S |f| \le |f(y)| + M - m$ for every $y \in S$. Taking the infimum over y gives $\sup_S |f| \le \inf_S |f| + M - m$. No maximum or minimum needs to be attained.

Conclusion. Rearrangement gives exactly the oscillation inequality needed in the proof of Theorem 33.5.

33.7 Exercise 33.7: Squaring preserves integrability

What we must show. Prove the Darboux-gap estimate and deduce integrability of f^2. Assume $|f(x)| \le B$ on $[a, b]$, with $B > 0$.

(a) Estimate the oscillation on one cell.

For any x, y in a partition cell J,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) + f(y)|\,|f(x) - f(y)| \le 2B(\sup_J f - \inf_J f).$$

Taking the supremum over pairs $x, y \in J$ gives

$$\sup_J f^2 - \inf_J f^2 \le 2B(\sup_J f - \inf_J f).$$

Indeed, the difference of a supremum and an infimum equals the supremum of all differences; the values need not be attained. Multiply by the length of J and sum over the partition to obtain

$$U(f^2, P) - L(f^2, P) \le 2B[U(f, P) - L(f, P)].$$

(b) Use the integrability criterion.

If f is integrable, choose P with $U(f, P) - L(f, P) < \varepsilon/(2B)$. The estimate makes the gap for f^2 smaller than ε. Also f^2 is bounded by B^2, as required by the criterion.

Conclusion. f^2 is integrable whenever f is integrable.

33.8 Exercise 33.8: Products, maxima, and minima

What we must show. Prove integrability of three combinations of integrable f, g.

(a) Turn a product into a linear combination of squares.

Linearity already proves that $f + g$ and $f - g$ are integrable. By Exercise 33.7 their squares are integrable. Since

$$fg = \frac{(f + g)^2 - (f - g)^2}{4},$$

linearity proves integrability of fg. This argument establishes the existence of the integral, without claiming that an integral of a product is a product of integrals.

(b) Express maxima and minima using absolute value.

For any two real numbers u, v, $\max(u,v) = (u+v+|u-v|)/2$ and $\min(u,v) = (u+v-|u-v|)/2$. Check these identities separately when $u \geq v$ and $u < v$. Substituting $f(x), g(x)$ gives pointwise formulas. The absolute value of the integrable function $f-g$ is integrable by Theorem 33.5, and linearity now proves both conclusions.

Conclusion. The functions fg, $\max(f,g)$, and $\min(f,g)$ are integrable.

33.9 Exercise 33.9: Uniform limits of integrable functions

What we must show. Show that a uniform limit f is integrable and its integral is the limit of the integrals.

First check boundedness.

Choose one index m such that $|f - f_m| < 1$ everywhere. Since f_m is bounded, this also bounds f.

Make a Darboux gap small.

Given $\varepsilon > 0$, choose m with $\sup_{[a,b]} |f - f_m| < \varepsilon/[4(b-a)]$. Call this upper bound η. For every cell J,

$$\sup_J f \leq \sup_J f_m + \eta, \qquad \inf_J f \geq \inf_J f_m - \eta.$$

Since f_m is integrable, choose P with its gap below $\varepsilon/2$. Then

$$U(f,P) - L(f,P) \leq U(f_m,P) - L(f_m,P) + 2\eta(b-a) < \varepsilon.$$

The Darboux criterion proves integrability of f.

Estimate the difference of integrals.

Now $f - f_n$ is integrable, and

$$\left| \int_a^b f - \int_a^b f_n \right| \leq \int_a^b |f - f_n| \leq (b-a) \sup_{[a,b]} |f - f_n| \longrightarrow 0.$$

Uniform convergence is what makes the final supremum tend to zero. Theorem 25.2 treated continuous functions; this proof needs only integrability. The dominated-convergence result has weaker convergence hypotheses but additional boundedness and limit-integrability assumptions.

Conclusion. Uniform limits preserve Riemann integrability and allow the limit to pass through the integral.

33.10 Exercise 33.10: An oscillatory function that is still integrable

What we must show. Prove integrability on $[-1,1]$ of $f(x) = \sin(1/x)$ for $x \neq 0$, with $f(0) = 0$.

Plan. Isolate the only troublesome point in a short interval. Given $\varepsilon > 0$, choose $0 < \delta < 1$ with $4\delta < \varepsilon/2$. The function is continuous on $[-1, -\delta]$ and $[\delta, 1]$, so those intervals have partitions whose

combined Darboux gaps are less than $\varepsilon/2$. Combine these partitions with the middle cell $[-\delta, \delta]$. On that cell $-1 \le f \le 1$, so its gap contribution is at most $2(2\delta) = 4\delta < \varepsilon/2$. The complete partition therefore has gap below ε.

Conclusion. The function is Riemann integrable. Infinitely many oscillations near one point can be confined to intervals of arbitrarily small length.

33.11 Exercise 33.11: An integrable function beyond the piecewise criteria

What we must show. Study $f(x) = x\,\mathrm{sgn}(\sin(1/x))$ for $x \neq 0$ and $f(0) = 0$, with $\mathrm{sgn}(0) = 0$.

(a) Rule out piecewise continuity.

At $x_k = 1/(k\pi) > 0$, the sine changes sign and the two one-sided limits of f are x_k and $-x_k$, in one order or the other. They differ because $x_k > 0$, so each x_k is a discontinuity. Every interval $(0, \delta)$ contains infinitely many of these points. A finite partition always leaves an interval $(0, \delta)$ without any interior partition point, and continuity on its interior is impossible. Thus f is not piecewise continuous on $[-1, 1]$.

(b) Rule out piecewise monotonicity.

The sign of $\sin(1/x)$ alternates on the infinitely many successive intervals between the x_k. In any $(0, \delta)$ choose ordered points $u < v < w$ in three consecutive sign intervals with signs $+, -, +$. Then $f(u) > 0$, $f(v) < 0$, and $f(w) > 0$. These values are neither nondecreasing nor nonincreasing. Hence f is not monotonic on any $(0, \delta)$, which again rules out any finite monotonicity partition.

(c) Estimate the Darboux gap directly.

Given $\varepsilon > 0$, choose $\delta \in (0, 1)$ with $4\delta^2 < \varepsilon/2$. On each of $[-1, -\delta]$ and $[\delta, 1]$ there are only finitely many zeros of $\sin(1/x)$. Between them the function equals x or $-x$; it is thus a finite piecewise continuous function, hence integrable there. Choose outer partitions with combined gaps below $\varepsilon/2$. On $[-\delta, \delta]$, $|f| \le |x| \le \delta$, so the gap contribution is at most $(2\delta)(2\delta) = 4\delta^2$. The combined gap is below ε.

Conclusion. f is neither piecewise continuous nor piecewise monotonic, but it is Riemann integrable. The two piecewise conditions are sufficient, not necessary.

33.12 Exercise 33.12: The rational-denominator function has zero integral

What we must show. For the function of Exercise 17.14, prove the two non-piecewise claims and compute the integral on every $[a, b]$. Recall that $f = 0$ at irrationals and $f(p/q) = 1/q$ for rationals in lowest terms with $q > 0$, including $f(0) = 1$.

(a) Examine every nondegenerate subinterval.

The function is discontinuous at every rational point: its value there is positive, whereas nearby irrational values are zero. Every nondegenerate interval contains a rational, so it cannot be an interval of continuity. Thus no finite continuity partition exists. It cannot be monotonic on any such interval either: choose a rational r strictly inside it and irrational points $u < r < v$ in it. Then $f(u) = f(v) = 0 < f(r)$, violating both possible monotonicities. Thus a finite monotonicity partition is also impossible.

(b) Isolate the finitely many large values.

Every cell contains an irrational, so every lower sum is zero. To make an upper sum small, let $\varepsilon > 0$ and choose N with $(b-a)/N < \varepsilon/2$. There are only finitely many rationals in $[a, b]$ with denominator $q \le N$: for each fixed q, only finitely many integers p have $a \le p/q \le b$. Make a partition whose cells meeting these finitely many points have total length less than $\varepsilon/2$. On these cells, use the bound $f \le 1$. On all other cells, all nonzero values have denominator $q > N$, and hence $f \le 1/N$. Therefore

$$U(f, P) \le 1 \cdot \frac{\varepsilon}{2} + \frac{1}{N}(b-a) < \varepsilon.$$

This proves the upper integral is also zero.

Conclusion. f is integrable on every bounded closed interval and $\int_a^b f = 0$, although it is neither of the two piecewise types.

33.13 Exercise 33.13: Equal integrals force an interior intersection

What we must show. For continuous f, g with equal integrals on $[a, b]$, find $x \in (a, b)$ with $f(x) = g(x)$. Let $h = f - g$. It is continuous and $\int_a^b h = 0$. Suppose h never vanishes in (a, b). By the Intermediate Value theorem it cannot take both a positive and a negative value there, since that would force a zero between them. Thus it is either positive everywhere in (a, b) or negative everywhere there. In the positive case, continuity gives $h \ge 0$ at the endpoints as well. At any chosen interior point x_0, continuity gives a small interval of positive length on which $h \ge h(x_0)/2 > 0$. Its integral is strictly positive, and the rest is nonnegative, contradicting the zero integral. Apply the same reasoning to $-h$ in the negative case.

Conclusion. The assumption was impossible, so $f(x) = g(x)$ for some interior x.

33.14 Exercise 33.14: A weighted mean-value theorem

What we must show. Prove the weighted formula, identify the ordinary case, and test the sign hypothesis.

(a) Separate the zero-weight case.

Let $G = \int_a^b g \ge 0$. If $G = 0$, nonnegative continuous g must be identically zero: a positive value would, by continuity, contribute a positive integral on a short interval. Thus $\int_a^b fg = 0$ too, and every interior x satisfies the formula. If $G > 0$, put $c = G^{-1} \int_a^b fg$. We need $f(x) = c$ at an interior point. Suppose no such point exists. As in Exercise 33.13, continuity forces $f - c$ to have one strict sign throughout (a, b). Because $G > 0$, g is positive at some interior point t_0; otherwise it would vanish on (a, b) and hence also at both endpoints. Thus $(f - c)g$ has one weak sign throughout $[a, b]$ and the corresponding strict sign on a neighborhood of t_0. Its integral cannot be zero. But linearity gives

$$\int_a^b (f - c)g = \int_a^b fg - cG = 0,$$

a contradiction. Therefore an interior x has $f(x) = c$, and $\int_a^b fg = f(x) \int_a^b g$.

(b) Recover the ordinary integral mean-value theorem.

Take $g \equiv 1$, so $G = b - a$. The formula becomes $f(x) = (b-a)^{-1} \int_a^b f$, exactly Theorem 33.9.

(c) Test a weight that changes sign.

On $[-1,1]$, set $f(t)=g(t)=t$. Then $\int_{-1}^{1} fg=\int_{-1}^{1} t^2\,dt=2/3$, but $f(x)\int_{-1}^{1} g=x\cdot 0=0$ for every x.

Conclusion. The weighted theorem holds for nonnegative continuous weights; part (c) shows that dropping the sign condition can destroy it.

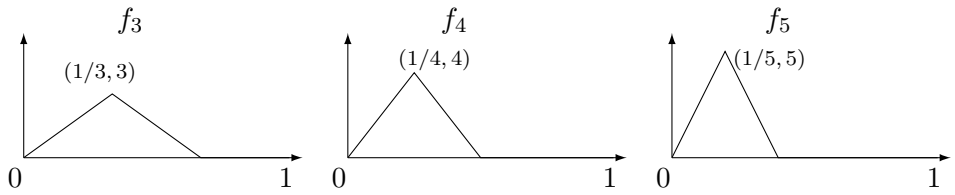
33.15 Exercise 33.15: Narrow triangular peaks

What we must show. Sketch the first three peaks, find their pointwise limit, and compare limits of integrals. The line-segment description gives the convenient formula

$$f_n(x)=\begin{cases} n^2x, & 0\le x\le 1/n,\\ 2n-n^2x, & 1/n\le x\le 2/n,\\ 0, & 2/n\le x\le 1. \end{cases}$$

(a) Draw the vertices and join them by straight segments.

The following sketches use the same horizontal and vertical scales; the vertical unit is 8 picture units and the horizontal unit is 100. The vertices are $(0,0),(1/n,n),(2/n,0),(1,0)$.



(b) Fix the point before choosing how large n must be.

At $x=0$, every value is zero. For a fixed $x>0$, choose $N>2/x$. If $n\ge N$, then $2/n<x$ and the last branch of the formula gives $f_n(x)=0$. Thus $f_n(x)\to 0$ for every $x\in[0,1]$.

(c) Compute areas and check the theorem's hypothesis.

Each graph is a triangle of base $2/n$ and height n, followed by a zero segment. Its integral is

$$\int_0^1 f_n(x)\,dx=\frac{1}{2}\frac{2}{n}\,n=1.$$

Hence the limit of the integrals is 1, whereas the integral of the pointwise limit is 0. There is no contradiction with Ross's Dominated Convergence theorem: it requires a single constant M bounding $|f_n(x)|$ for all n,x, but $f_n(1/n)=n$ is unbounded as n varies.

Conclusion. Pointwise convergence to zero does not force the integrals to tend to zero; the increasingly tall peaks retain area 1.